50.007 Machine Learning

Ensemble Models

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Learning Objectives

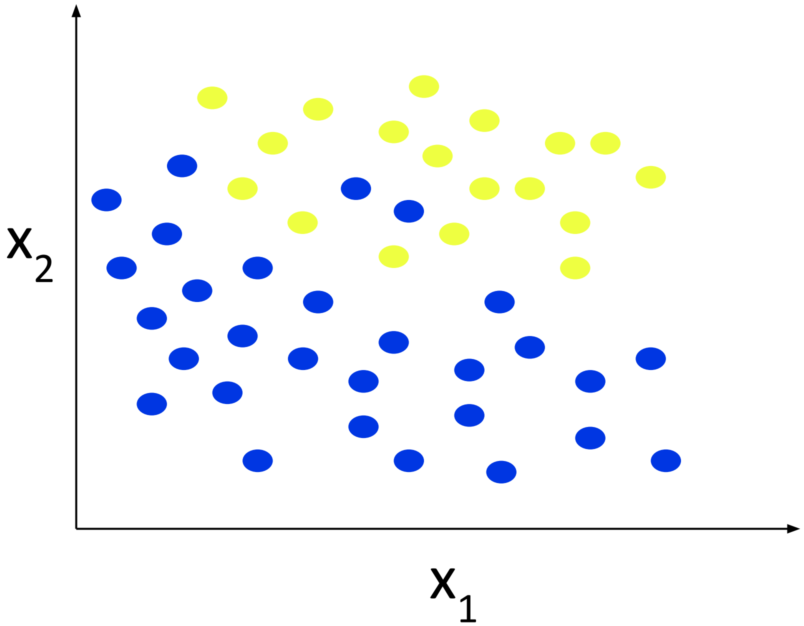
You should be able to know:

- What is Generalization? Underfitting and overfitting
- Types of ensemble classifiers
  - Bagging (Random Forest)
  - Boosting
- How to model evaluation

Generalization

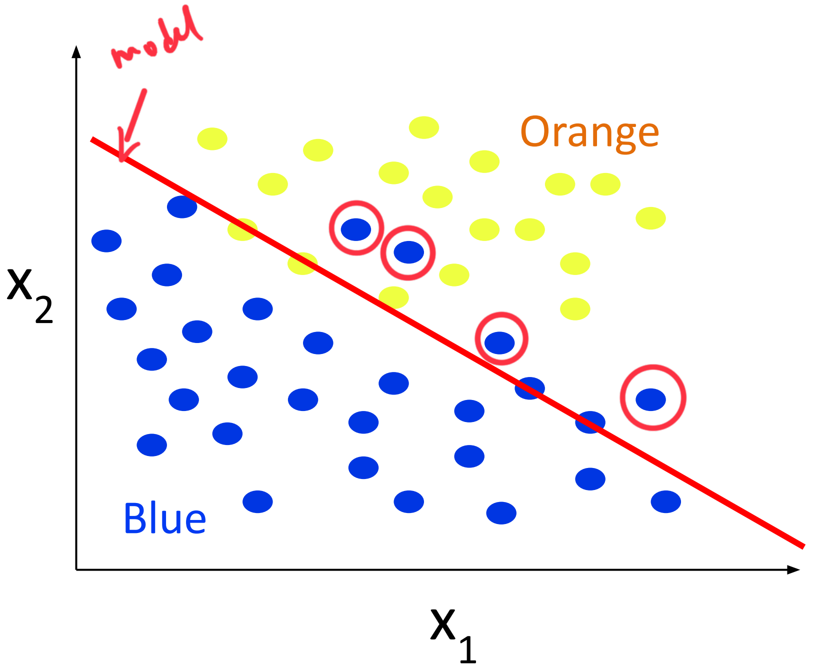
- Training data: $\{x_i, y_i\}$
  - Examples that we use to train our model
  - E.g. past records of days labelled with Roy goes/not go fishing
- Future/test data: $\{x_i, \hat{y}_i\}$
  - Examples that our model has not seen before
  - E.g. information about tomorrow to predict if Roy goes fishing
- Want to do well on future data not training data! $\longrightarrow$ for future unseen data.
  - Not very useful to do well on training data; we already know the label
  - Easy to be perfect on training data
  - Doing well in training does not mean will do well on future data!

Classification Example



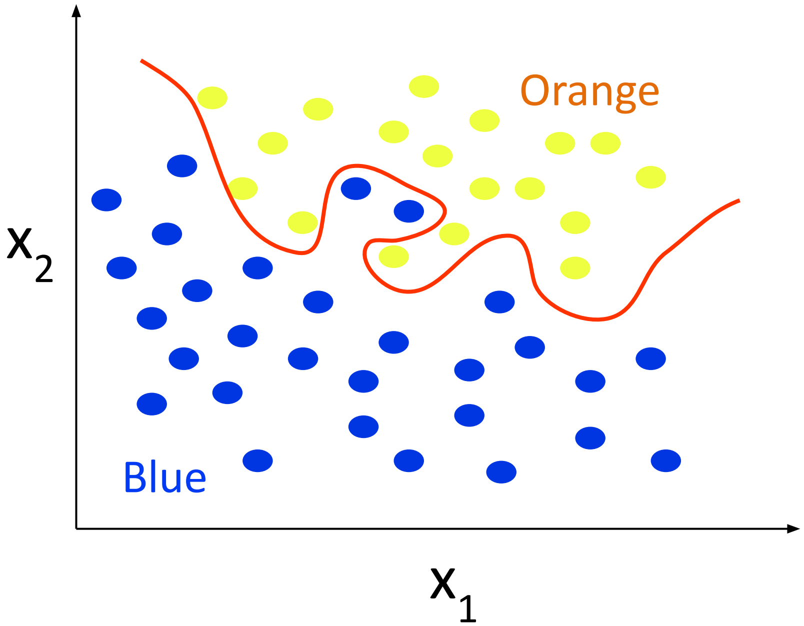
- Find a way to classify the points

Classification Example



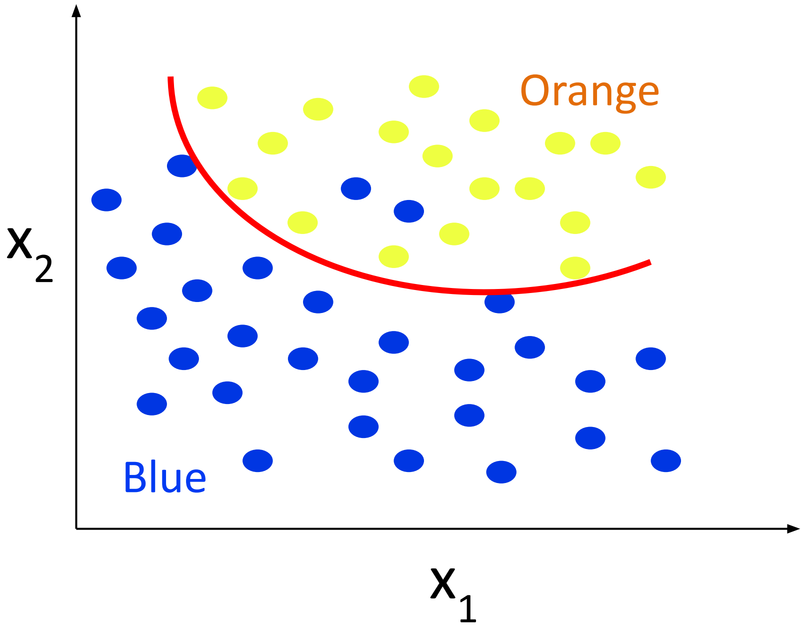
- Simple solution
- Unable to capture all salient pattern in data

Classification Example



- Complex solution
- "Try too hard" to work well with training data
- Fit noise into the model
- Pattern is one-off and may not appear again in future data

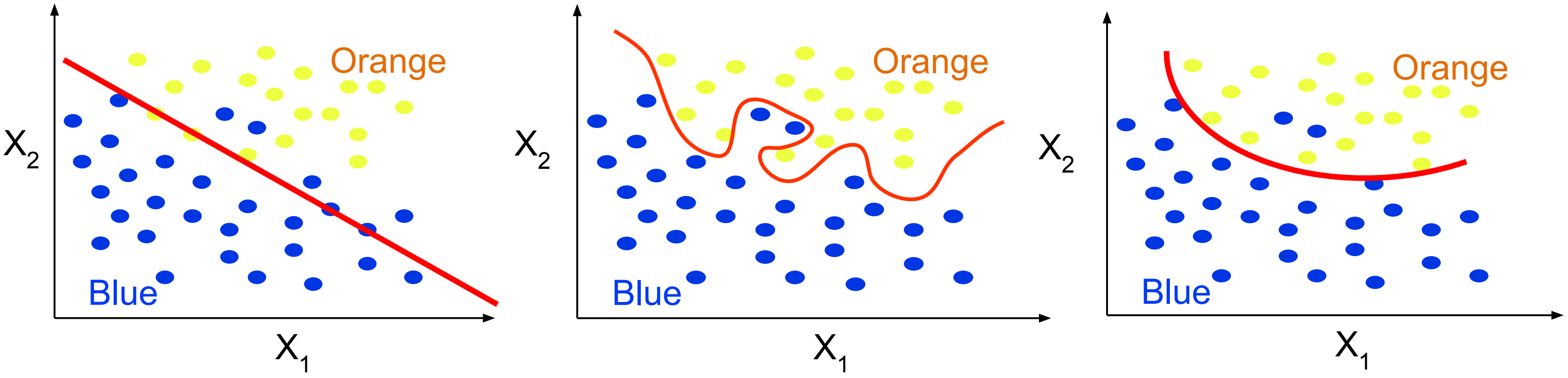
Classification Example



- Good solution, able to generalize well for future data

but do not overfit !

Underfitting and Overfitting



Underfitting
(Too simple to explain the variance)

Overfitting
(Too complex - try too hard)

Appropriate Fitting

Underfitting and Overfitting

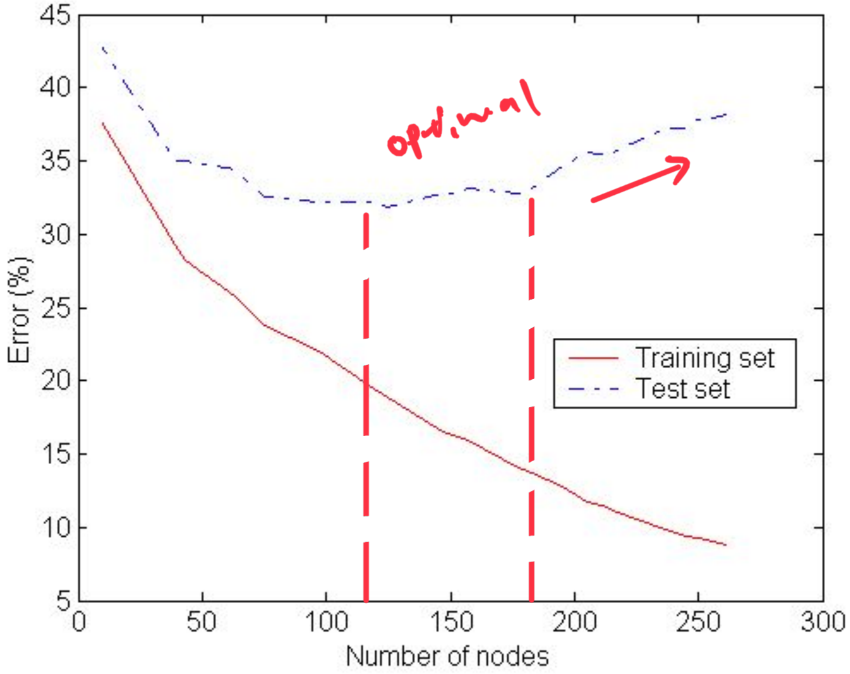
- **Overfitting**
  - Model is too complex (too flexible)
  - Fit noise in training data
  - Pattern is one-off and might not appear in future data
  - Model F overfits the data if:
    - We can find another model F’
    - Which makes more mistakes in training data: $E_{Train}(F') > E_{Train}(F)$
    - But less mistake in future data: $E_{Test}(F) > E_{Test}(F')$
- **Underfitting**
  - Model is too simple
  - Not powerful enough to capture the salient pattern
  - Can find another model F’ with smaller $E_{Test}(F')$ and $E_{Train}(F')$

intuition

Error in model F' on training

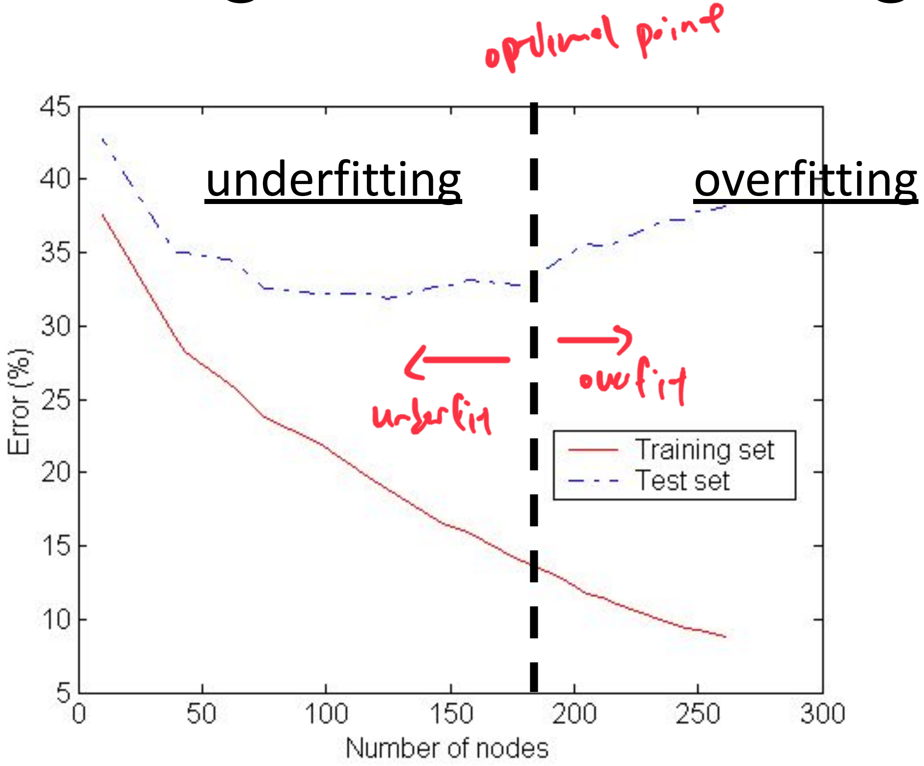
Error in model F on training

Underfitting and Overfitting



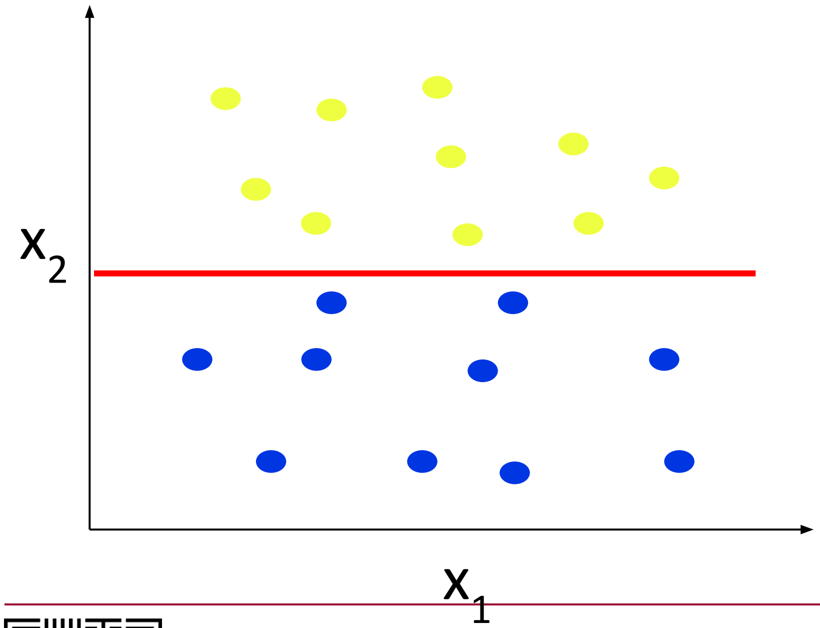
As you ↑ the # of nodes,
the % error ↑ as well.

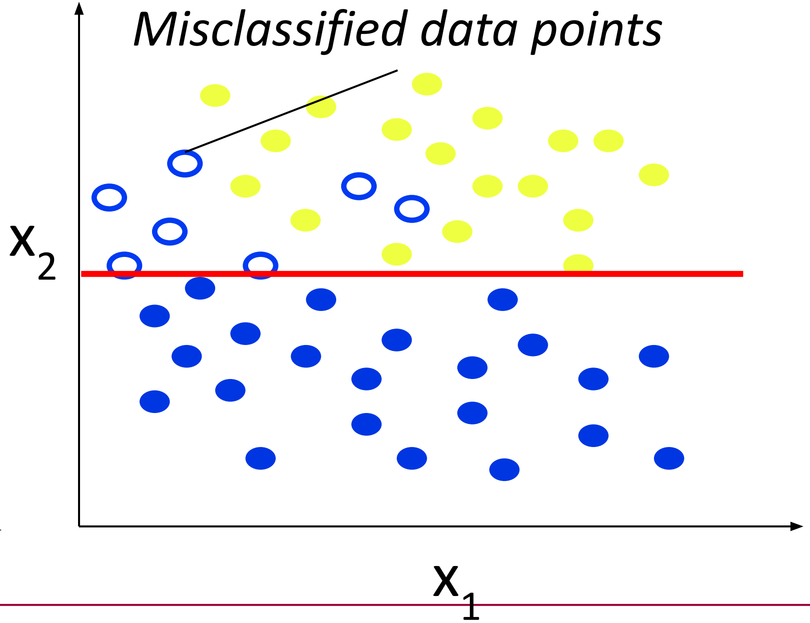
Underfitting and Overfitting



What Causes Overfitting?

- Noise in training data
- Insufficient data points in training data





Further Notes on Overfitting

- Overfitting results in decision trees that are more complex than necessary

- Training error no longer provides a good estimate of how well the tree will perform on previously unseen records

- Need new ways for estimating errors

Estimating Generalization Errors

*IMPORTANT
MCQ / T&F

- Training errors: error on training (Σ e(t))
- Generalization errors: error on testing data (Σ e'(t))
- Methods for estimating generalization errors:
  - 1 Optimistic approach: simply use training error
    - e'(t) = e(t) ⇒ Error on training = Error on testing
  - 2 Pessimistic approach:
    - For each leaf node: e'(t) = (e(t)+0.5) → arbitrary / estimated.
    - Total errors: e'(T) = e(T) + N × 0.5 (N: number of leaf nodes)
    - Example: For a tree with 30 leaf nodes and 10 errors on training data (out of 1000 training data instances):
      - 1 Training error = 10/1000 = 1%
      - 2 Generalization error = (10 + 30×0.5)/1000 = 2.5% → Estimation
  - Reduced error pruning (REP):
    - uses validation data set to estimate generalization error

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Estimating Generalization Errors

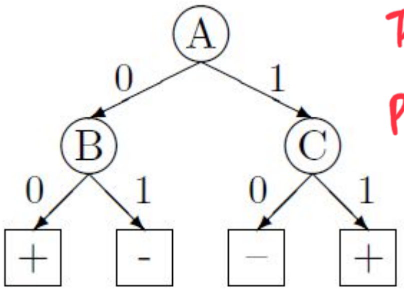
Training

S/N	A	B	C	Label
1	0	0	0	+
2	0	0	1	+
3	0	1	0	+
4	0	1	1	-
5	1	0	0	+

Testing

S/N	A	B	C	Label
6	0	0	0	+
7	0	1	1	+
8	1	1	0	+
9	1	0	1	-
10	1	0	0	+

model



Training Error $= \frac{2}{5} = 0.4$

Pessimistic Error:

$= (2 + 4 \times 0.5)/5$

$= 0.8$

Actual $= \sum e'(t)$

$= 4/5 = 0.8$

Estimating Generalization Error

Optimistic: ? $e(t) = e'(t) = \frac{2}{5} = 0.4$

Pessimistic: ? $e'(t) = e(t) + N \times 0.5$

$= (2 + 4 \times 0.5)/5 = 0.8$

Actual Generalization Error : ?

$\sum e'(t) = \frac{4}{5} = 0.8$

Estimating Generalization Errors

Training

S/N	A	B	C	Label	
1	0	0	0	+	✓
2	0	0	1	+	✓
3	0	1	0	+	✗
4	0	1	1	-	✓
5	1	0	0	+	✗

Testing

S/N	A	B	C	Label	
6	0	0	0	+	✓
7	0	1	1	+	✗
8	1	1	0	+	✗
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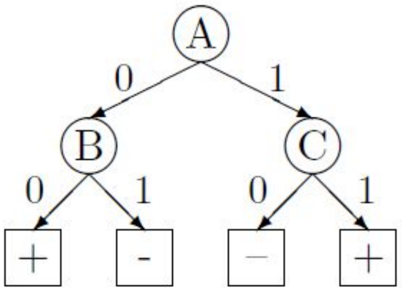

Estimating Generalization Error

Optimistic: ?

Pessimistic: ?

Actual Generalization Error : ?

Estimating Generalization Errors

Training

S/N	A	B	C	Label	
1	0	0	0	+	✓
2	0	0	1	+	✓
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Testing

S/N	A	B	C	Label	
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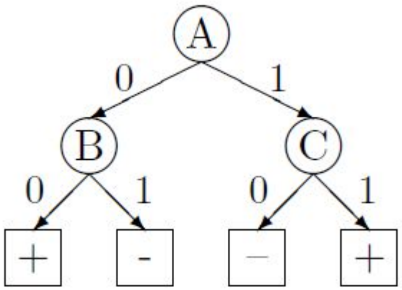

Estimating Generalization Error

normalising

Optimistic: e'(t) = e(t) = 2 / 5 = 0.4

Pessimistic: e'(t) = (e(t)+0.5) = (2 + 4 * 0.5)/5
= 0.8

Actual Generalization Error : e'(t) = 4 / 5 = 0.8

Occam's Razor

- Given two models of similar generalization errors, one should prefer the simpler model over the more complex model

Big-O Complexity

- For complex models, there is a greater chance that it was fitted accidentally by errors in data

- Therefore, one should include model complexity when evaluating a model

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How to Address Overfitting

- **Pre-Pruning (Early Stopping Rule)**
  - Stop the algorithm before it becomes a fully-grown tree
  - Typical stopping conditions for a node:
    - Stop if all instances belong to the same class
    - Stop if all the attribute values are the same
    - More restrictive conditions: E.g., Stop if expanding the current node does not improve impurity measures (e.g., Gini or information gain).
- **Post-Pruning**
  - Grow decision tree to its entirety
  - Trim the nodes of the decision tree in a bottom-up fashion
    - If generalization improves after trimming, replace sub-tree by a leaf node

Performance of test improve

  - Class label of leaf node is determined from majority class of instances in the sub-tree

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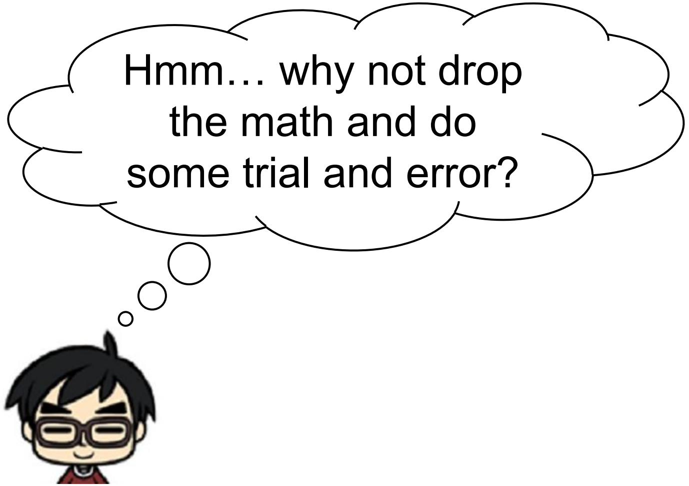
How will lazy Roy do it?



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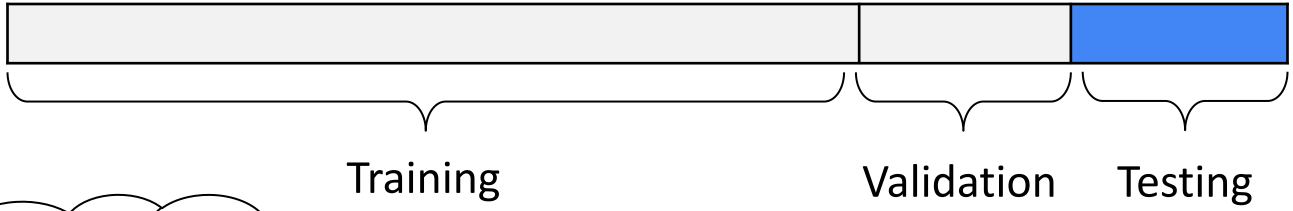
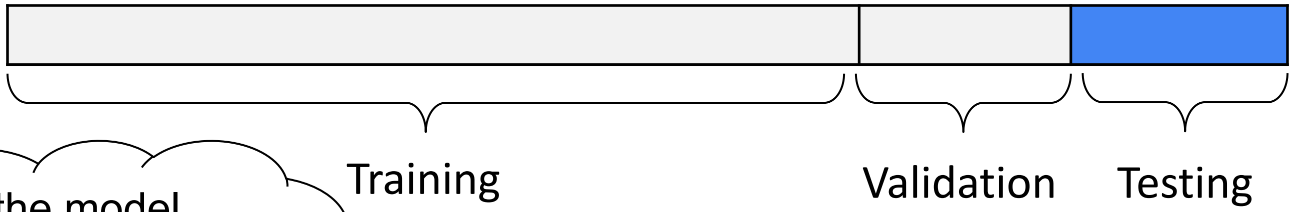


defined by user

First split the dataset into 3 parts…

How will lazy Roy do it?



We train the model using the training data and test it on validation set…

```
DT = DecisionTreeClassifier(depth=d)
DT.fit(training data)
Pred = DT.predict(validation data)
Acc = ComputeAcc(Pred)
```

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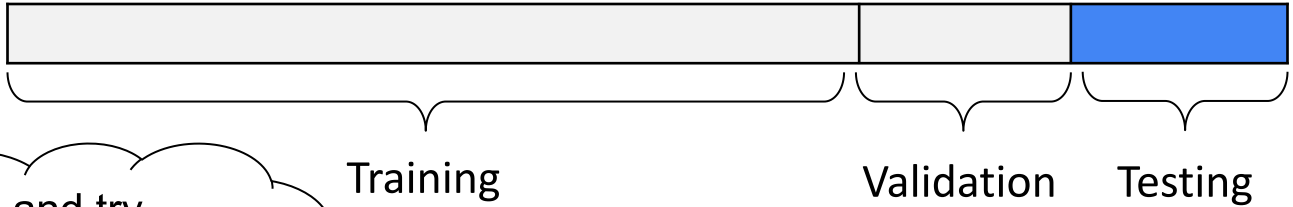

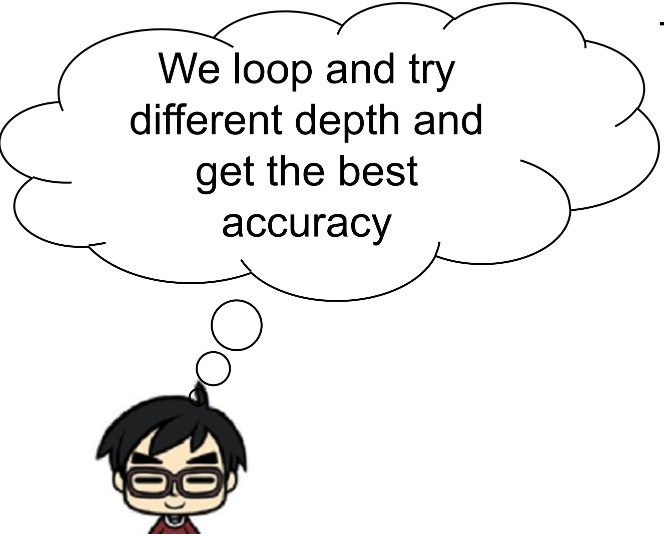


```
For d in range(101,1,-1)
    DT = DecisionTreeClassifier(depth=d)
    DT.fit(training data)
    Pred = DT.predict(validation data)
    Acc = ComputeAcc(Pred)
    if BestAcc < Acc:
        BestAcc = Acc
        BestDepth = d
```

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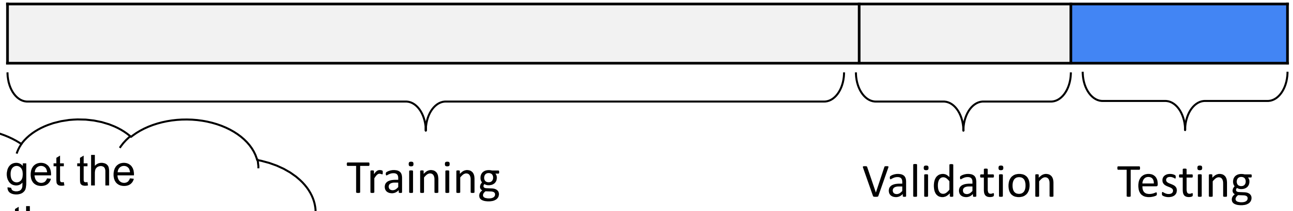

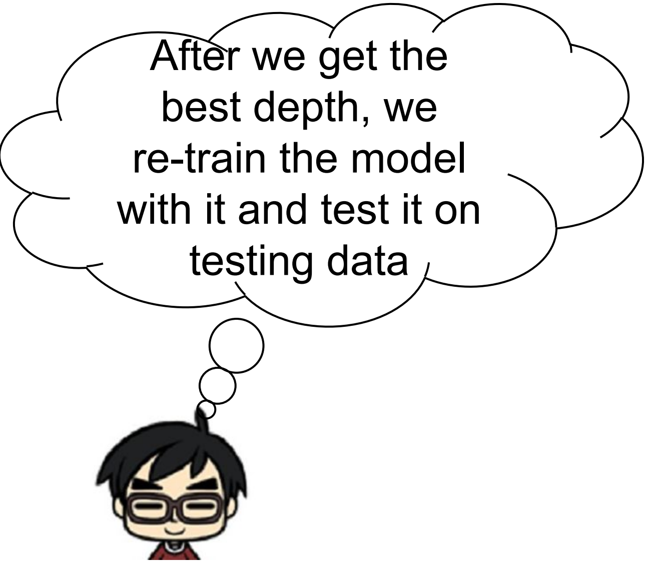


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     if BestAcc < Acc:
          BestAcc = Acc
          BestDepth = d
DT = DecisionTreeClassifier(depth=BestDepth)
DT.fit(training data) → training + validation
Pred = DT.predict(testing data)
Acc = ComputeAcc(Pred)
```

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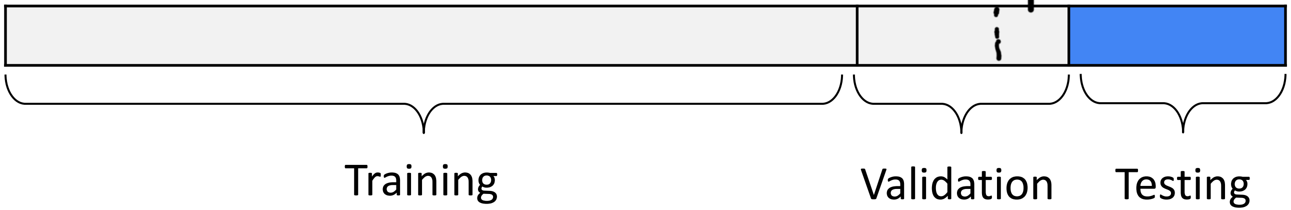
How will lazy Roy do it?

* For project, might need to cut from validation to test ∵ testing data have no labels.

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DT.fit(training data)
Pred = DT.predict(testing data)
Acc = ComputeAcc(Pred)
```

Caveat: workable solution but still have some drawbacks…

It will take too slow!!!

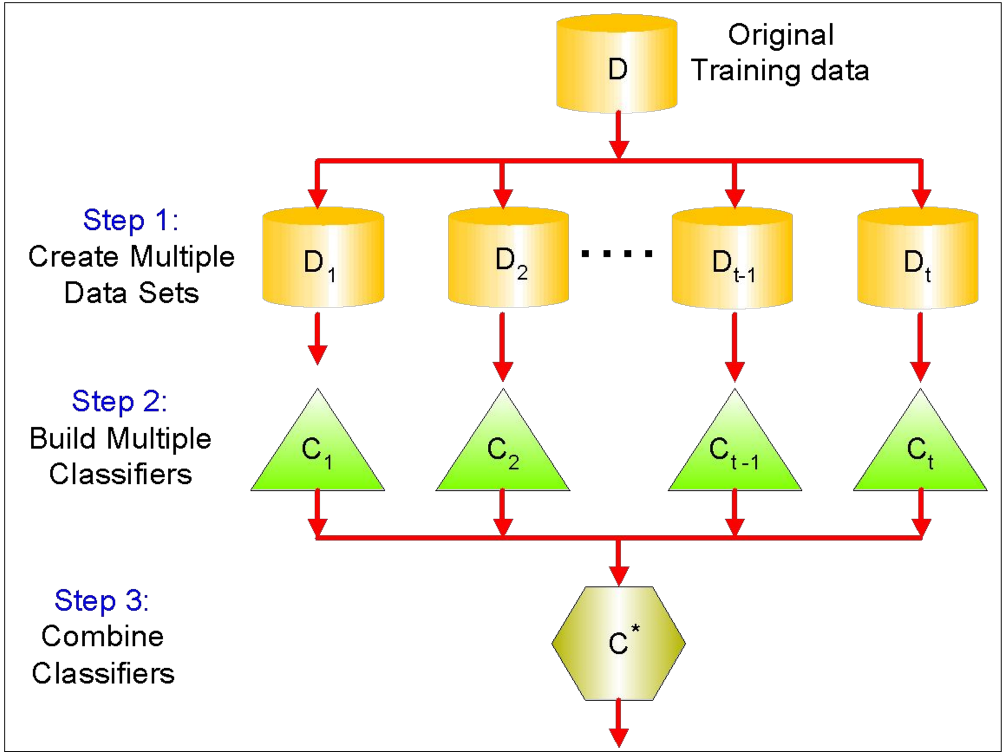
Ensemble Methods ☆ IMPORTANT

- Construct a set of classifiers from the training data

- Predict class label of previously unseen records by aggregating predictions made by multiple classifiers

- Assumption:
  - Individual classifiers (voters) could be lousy (stupid), but the aggregate (electorate) can usually classify (decide) correctly.

General Idea



t = usually odd #
to ensure somebody wins.

Why does it work?

- Suppose there are 25 base classifiers
  - Each classifier has error rate, $\varepsilon = 0.35$
  - Assume classifiers are independent
  - Probability that the ensemble classifier makes a wrong prediction (i.e.,*13 out of the 25 classifiers misclassified*):

$$\sum_{i=13}^{25} \binom{25}{i} \varepsilon^{i} (1-\varepsilon)^{25-i} = 0.06$$

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Ensemble Methods

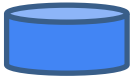
- How to generate an ensemble of classifiers?
  (list below is non exhaustive!)
  - Bagging
  - Boosting

Bagging

- **B**ootstrap **Agg**regating (Bagging)
- Multiple models of same learning algorithm trained with subset of dataset randomly sampled (with replacement) from the training dataset
- Each sample has probability $1-(1 – 1/n)^n$ of being selected
  - This value tends to be **0.63** for large n

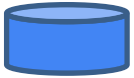
due to the central limit theorem (CLT)
∵ if n is large, the P falls to a bell curve,
there is a ↑ P ≈ 0.63

Bagging



Training Dataset
(Size *N*)

Bagging



Training Dataset
(Size *N*)

- Randomly select data points into the bags (repetition allow)
- Each bag has M data points, M<N

Bagging

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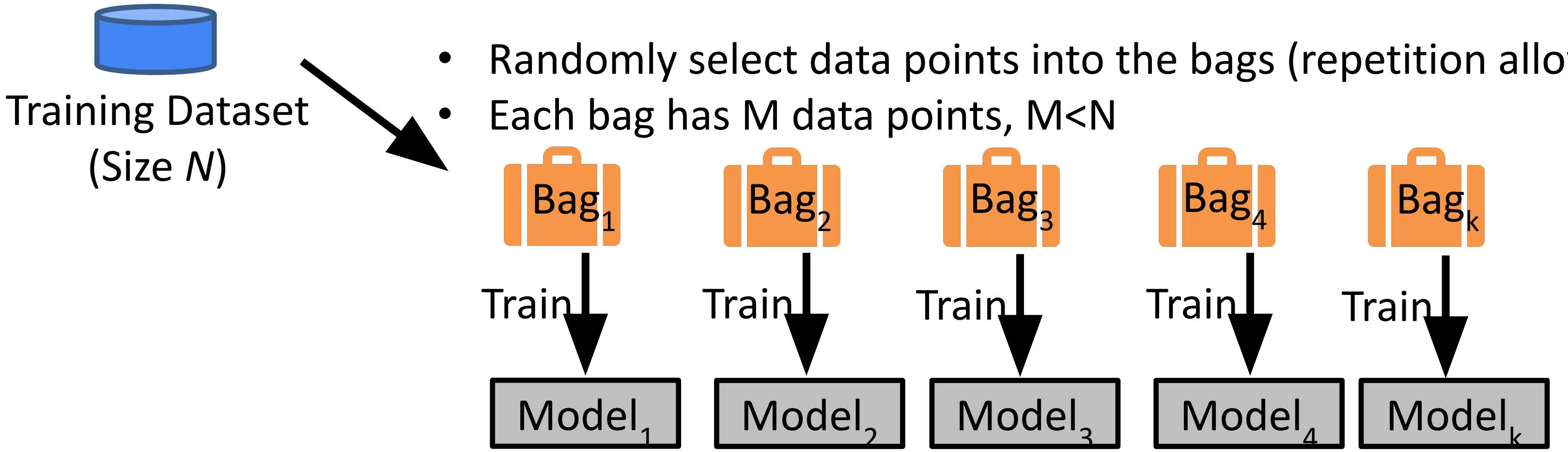

Bagging

IMPORTANT

Training Dataset
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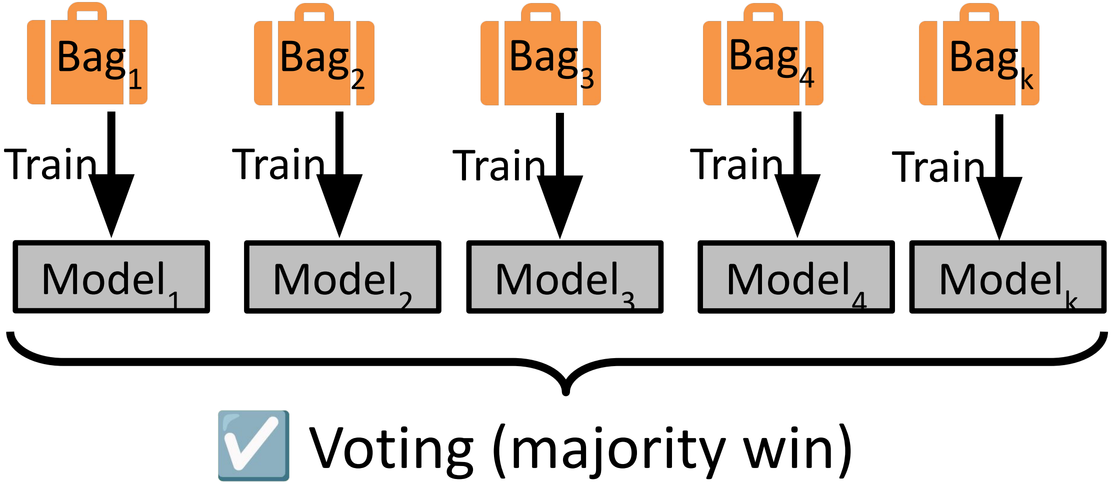

Random Forest

- Train a lot of decision trees and use bagging
- Wisdom of the crowd!
    - Uncorrelated trees are preferred

Random Forest

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- Wisdom of the crowd!
    - Uncorrelated trees are preferred

Random Forest

- Bootstrapping + Feature Randomness

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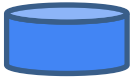
Boosting

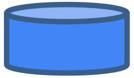
⇒ Analogy: For labels they guessed wrong, the model will go back & keep guessing till it is correct.

- An iterative procedure to adaptively change distribution of training data by focusing more on previously misclassified records
  - Initially, all N records are assigned equal weights
  - Unlike bagging, sampling weights may change at the end of boosting round
    - Records that are wrongly classified will have their weights increased
    - Records that are correctly classified will have their weights decreased
  - **Caveat**: boosting show better predictive accuracy than bagging but also tends to over-fits the training data

Boosting

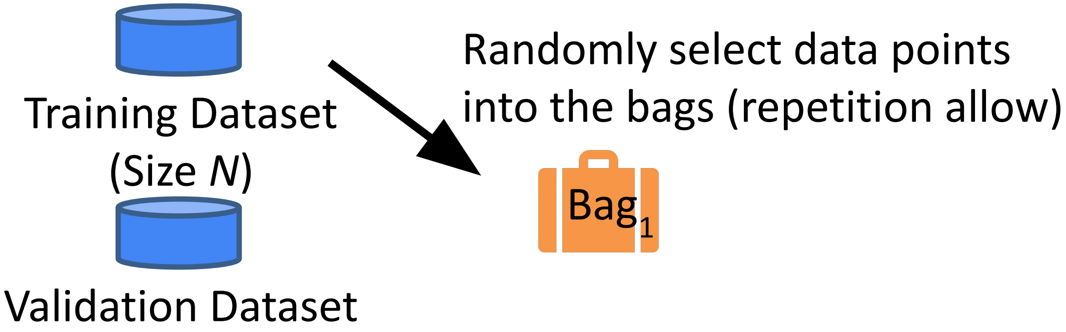


Training Dataset
(Size *N*)

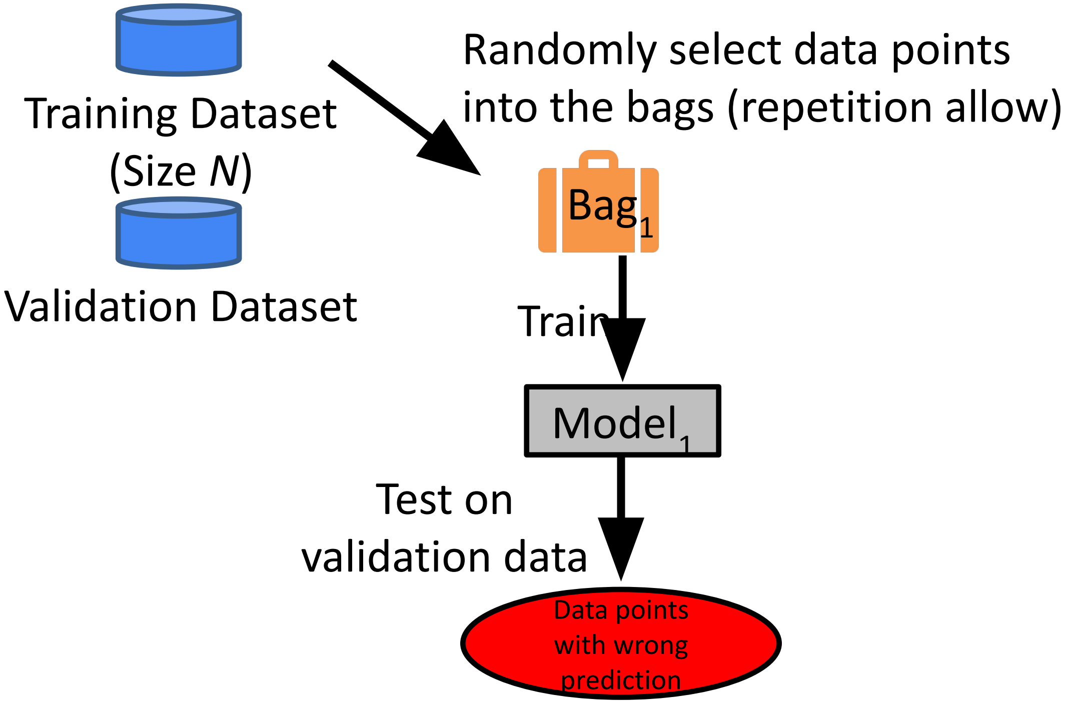


Validation Dataset

Boosting


Training Dataset
(Size N)
Validation Dataset
Randomly select data points
into the bags (repetition allow)
Bag_1

Boosting


Training Dataset
(Size N)
Validation Dataset
Randomly select data points
into the bags (repetition allow)
Bag1
Train
Model1
Test on
validation data
Data points
with wrong
prediction

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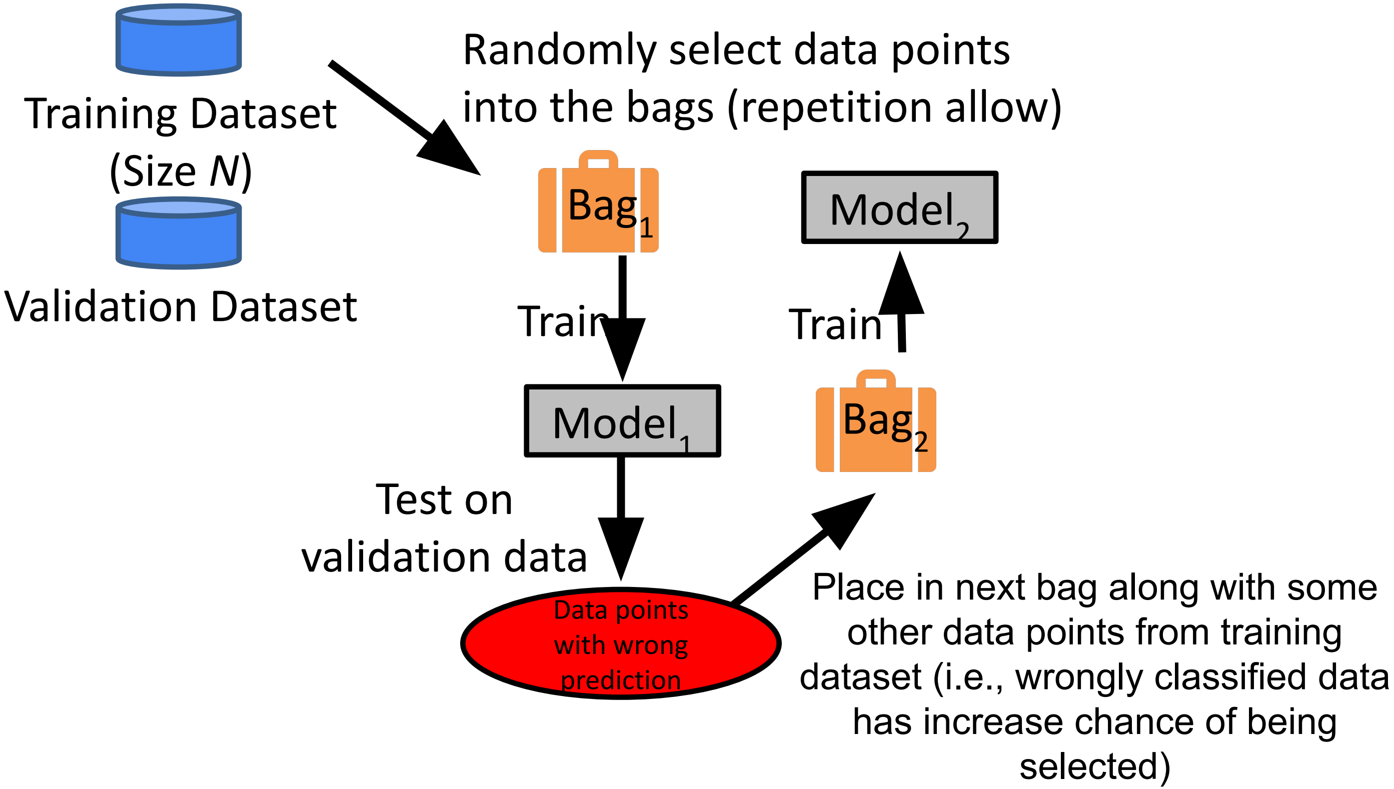
Boosting


Training Dataset
(Size N)
Validation Dataset
Randomly select data points
into the bags (repetition allow)
Bag1
Train
Model1
Test on
validation data
Data points
with wrong
prediction
Place in next bag along with some
other data points from training
dataset (i.e., wrongly classified data
has increase chance of being
selected)
Bag2
Train
Model2

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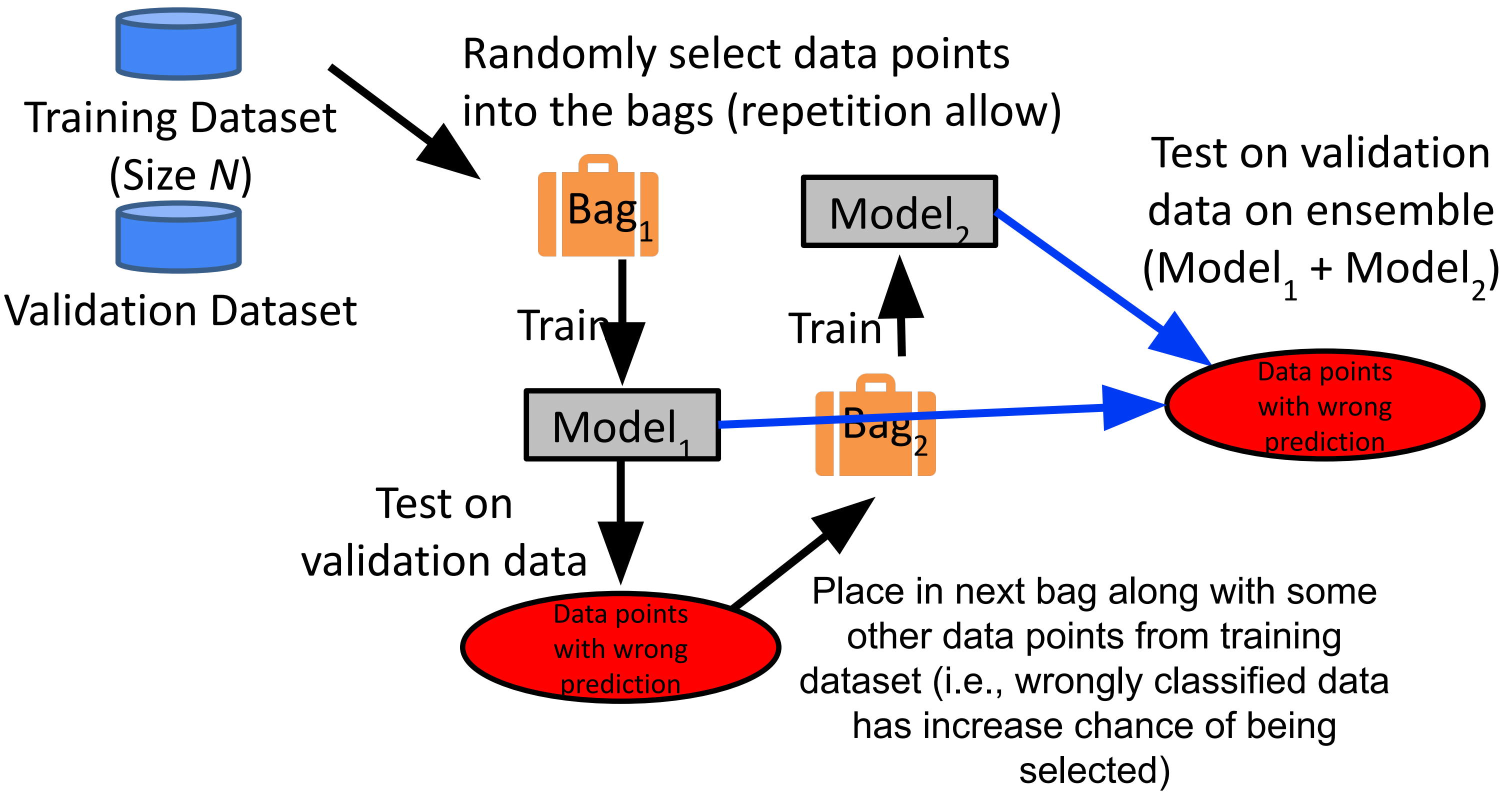
Boosting

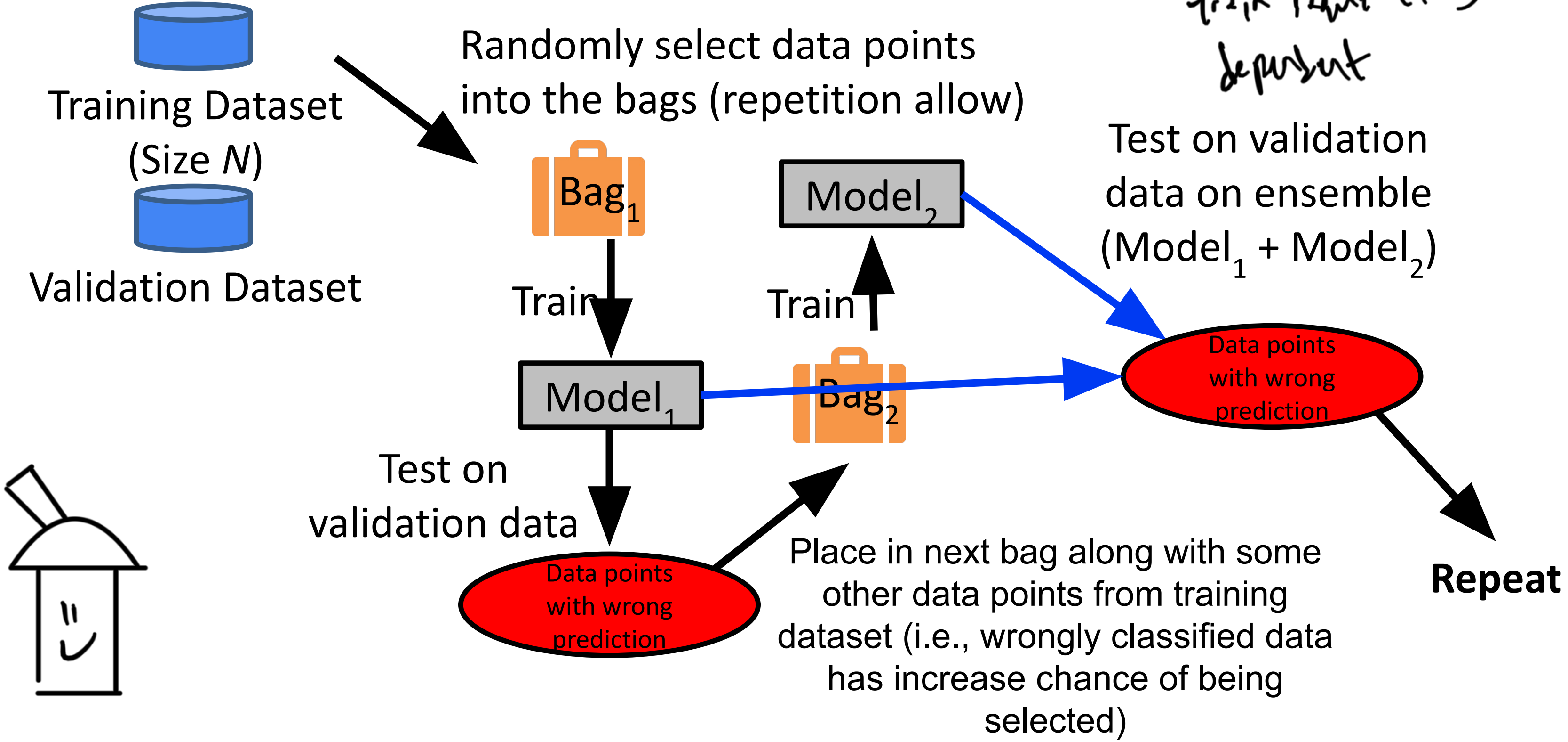


Training Dataset
(Size *N*)
Validation Dataset
Randomly select data points
into the bags (repetition allow)
Bag_1
Train
$Model_1$
Test on
validation data
Data points
with wrong
prediction
Place in next bag along with some
other data points from training
dataset (i.e., wrongly classified data
has increase chance of being
selected)
Bag_2
Train
$Model_2$
Test on validation
data on ensemble
($Model_1$ + $Model_2$)
Data points
with wrong
prediction

Boosting

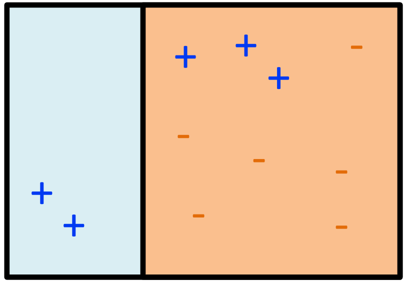
*IMPRTANT

* Cannot train while parallel, train sequentially ∵ dependent

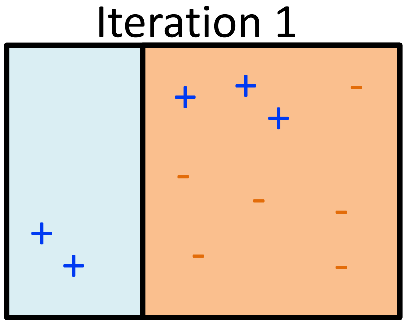


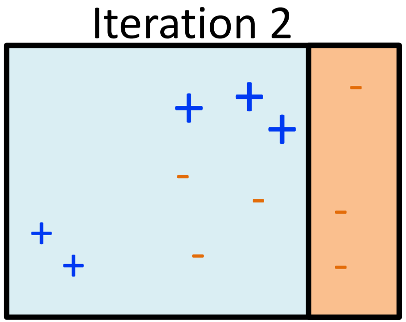
AdaBoost

Iteration 1

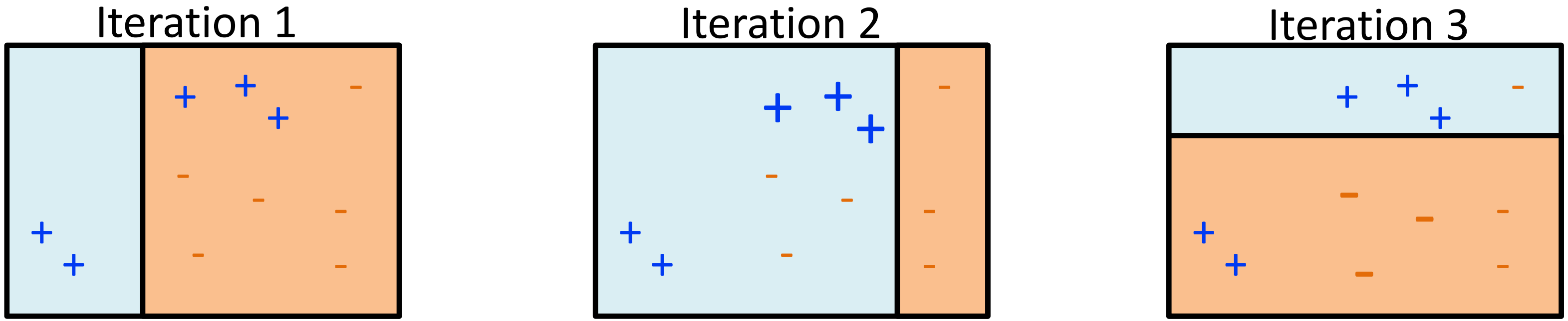


AdaBoost

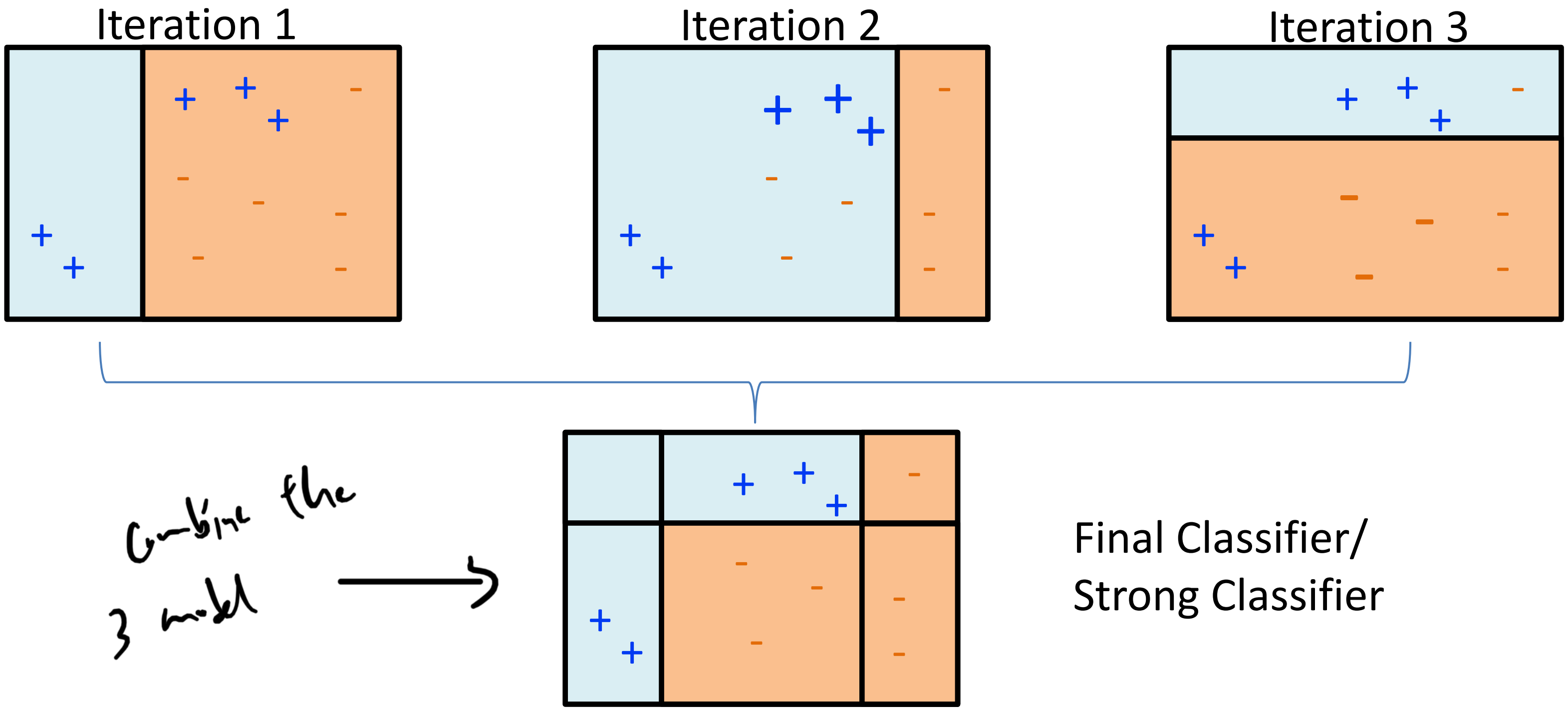




AdaBoost



AdaBoost


Iteration 1
Iteration 2
Iteration 3
Combine the 3 model
Final Classifier/
Strong Classifier

AdaBoost Algorithm

- Initialize observation weights $w(x_i, y_i) = 1/n,\ i=1,\ldots,n$
- Base classifiers: $C_1, C_2, \ldots, C_T$

equal chance

AdaBoost Algorithm

- Initialize observation weights $w(x_i, y_i) = 1/n,\ i=1,\ldots,n$
- Base classifiers: $C_1, C_2, \ldots, C_T$
- For $i=1$ to T:
    - Fit a classifier $C_i(x)$ to training data
    - Compute error rate of C_i:

$$\boxed{\varepsilon_i = \frac{1}{N}\sum_{j=1}^{N} w_j \delta\left(C_i(x_j) \neq y_j\right)}$$

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AdaBoost Algorithm

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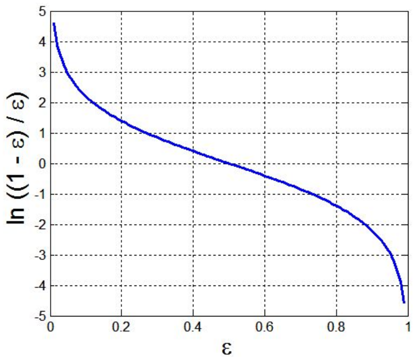
    - Fit a classifier $C_i(x)$ to training data
    - Compute error rate of C_i:

$$\boxed{\varepsilon_i = \frac{1}{N}\sum_{j=1}^{N} w_j \delta\left(C_i(x_j) \neq y_j\right)}$$

    - Compute importance of classifier C_i:

$$\boxed{\alpha_i = \frac{1}{2}\ln\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)}$$



When the error rate goes beyond 0.5, we assign the classifier a negative weight (i.e., we don't want them!)

$\varepsilon > 0.5$, worse than random, $\therefore$ importance < 0

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AdaBoost Algorithm

- Initialize observation weights $w(x_i,y_i) = 1/n,\ i=1,\ldots,n$
- Base classifiers: $C_1, C_2, \ldots, C_T$
- For $i=1$ to T:
  - Fit a classifier $C_i(x)$ to training data
  - Compute error rate of C_i:

$$\boxed{\varepsilon_i = \frac{1}{N}\sum_{j=1}^{N} w_j \delta\left(C_i(x_j) \neq y_j\right)}$$

  - Compute importance of classifier C_i :

$$\boxed{\alpha_i = \frac{1}{2}\ln\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)}$$

  - Update the weights

$$\boxed{\begin{array}{l} w_i^{(j+1)} = \dfrac{w_i^{(j)}}{Z_j}\begin{cases}\exp^{-\alpha_j} & \text{if } C_j(x_i) = y_i \\ \exp^{\alpha_j} & \text{if } C_j(x_i) \neq y_i\end{cases} \\ \text{where } Z_j \text{ is the normalization factor}\end{array}}$$

- $\exp^{-\alpha}$ is always lesser than 1 when the data point is correctly classified, thus giving it a smaller weight

- $\exp^{\alpha}$ is bigger than 1 when it is misclassified, assigning this data point a greater weight

AdaBoost Algorithm *IMPORTANT

- Initialize observation weights $w(x_i, y_i) = 1/n,\ i=1,\dots,n$
- Base classifiers: $C_1, C_2, \dots, C_T$
- For $i=1$ to T:
  - Fit a classifier $C_i(x)$ to training data
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  - Final Classifier:

    $$\boxed{C^*(x) = \arg\max_{y} \sum_{j=1}^{T} \alpha_j \delta\left(C_j(x) = y\right)}$$

Model Evaluation

- Metrics for Performance Evaluation
  - How to evaluate the performance of a model?
- Methods for Performance Evaluation
  - How to obtain reliable estimates?
- Methods for Model Comparison
  - How to compare the relative performance among competing models?

Metrics for Performance Evaluation

- Focus on the predictive capability of a model
  - Rather than how fast it takes to classify or build models, scalability, etc.
- Confusion Matrix:

	PREDICTED CLASS $\hat{y} = y'$	
	Class=Yes	Class=No
ACTUAL CLASS y — Class=Yes	a	b
ACTUAL CLASS y — Class=No	c	d

a: TP (true positive)

b: FN (false negative)

c: FP (false positive)

d: TN (true negative)

Metrics for Performance Evaluation

	PREDICTED CLASS		
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

Most widely-used metric:

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Limitation of Accuracy

- Consider a 2-class problem
  - Number of Class 0 examples = 9990
  - Number of Class 1 examples = 10

- If model predicts everything to be class 0, accuracy is 9990/10000 = 99.9 %
  - Accuracy is misleading because model does not detect any class 1 example

Cost Matrix *IMPORTANT MCQ / T&F

- $C(i|j)$: Cost of misclassifying class *j* example as class *i*

	PREDICTED CLASS		
ACTUAL CLASS	$C(i\|j)$	**Class=Yes**	**Class=No**
	Class=Yes	C(Yes|Yes) TP	C(No|Yes) FN
	Class=No	C(Yes|No) FP	C(No|No) TN

TP: True Positive

TN: True Negative

FN: False Negative

FP: False Positive

Computing Cost of Classification

Cost Matrix	PREDICTED CLASS		
ACTUAL CLASS	C(i|j)	+	-
	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	150	40
	-	60	250

Accuracy = ? $\frac{TP+TN}{Total} = \frac{150+250}{150+60+40+250} = \frac{400}{500} = \frac{4}{5} = 0.8$

Cost = ?

Model M_2	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	250	45
	-	5	200

Accuracy = ? $\frac{250+200}{250+5+45+200} = \frac{450}{500} = \frac{45}{50} = \frac{9}{10} = 0.9$

Cost = ?

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$$\text{cost} = 150(-1) + 40(100) + 60(1) + 250(0)$$
$$= -150 + 4000 + 60 + 0$$
$$= 3850 + 60$$
$$= 3910$$

$$\text{cost} = 250(-1) + 45(100) + 5(1) + 200(0)$$
$$= -250 + 4500 + 5 + 0$$
$$= 4250 + 5$$
$$= 4255$$

$\therefore$ M_2 might be have better accuracy but it has a higher cost.
M_1 might have a lower accuracy but much significant cost, $\therefore$ business might go with M_1 $\because$ of lower cost.

Computing Cost of Classification

Cost Matrix	PREDICTED CLASS		
ACTUAL CLASS	C(i|j)	+	-
	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	150	40
	-	60	250

Accuracy = 150 + 250/150+250+40+60

= 80%

Cost = 150(-1) + 40(100) + 60(1) + 250(0)

= 3910

Model M_2	PREDICTED CLASS		
ACTUAL CLASS		+	-
	+	250	45
	-	5	200

Accuracy = 250 + 200/250+200+45+5

= 90%

Cost = 250(-1) + 45(100) + 5(1) + 200(0)

= 4255

Precision, Recall & F1 * IMPORTANT

	PREDICTED CLASS	
	Class=Yes	Class=No
ACTUAL CLASS: Class=Yes	a (TP)	b (FN)
ACTUAL CLASS: Class=No	c (FP)	d (TN)

$$\text{Precision (p)} = \frac{a}{a+c} \qquad \frac{TP}{TP + FP}$$

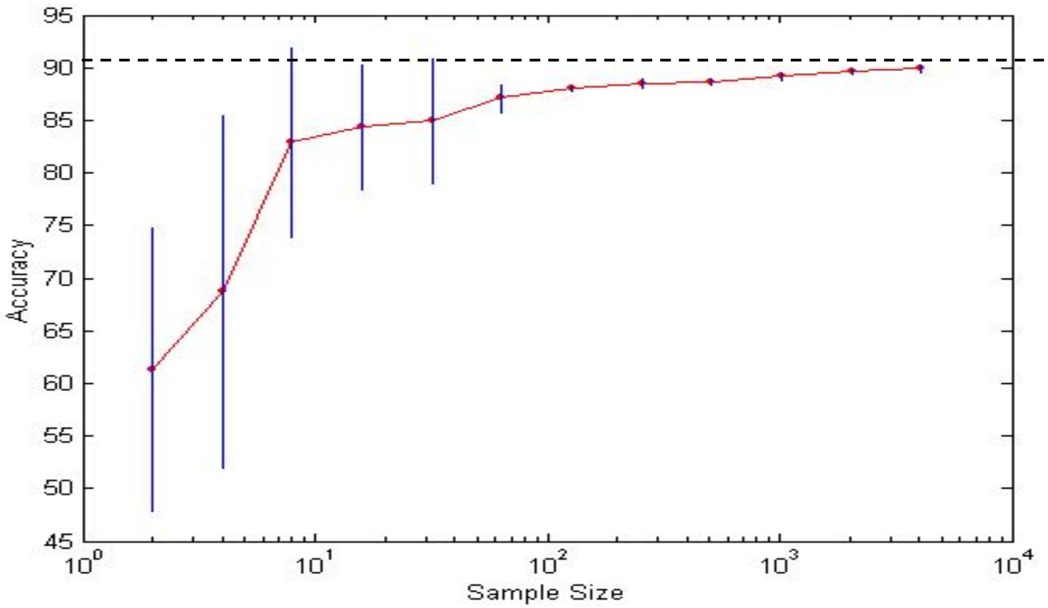
$$\text{Recall (r)} = \frac{a}{a+b} \qquad \frac{TP}{TP + FN}$$

$$\text{F - measure (F)} = \frac{2rp}{r+p} = \frac{2a}{2a+b+c}$$

Methods for Performance Evaluation

- How to obtain a reliable estimate of performance?
- Performance of a model may depend on other factors besides the learning algorithm:
  - Class distribution
  - Cost of misclassification
  - Size of training and test sets

Learning Curve



- Learning curve shows how accuracy changes with varying sample size
- Requires a sampling schedule for creating learning curve:
  - Arithmetic sampling (Langley, et al.)
  - Geometric sampling (Provost et al,)
- Effect of small sample size:
  - Bias in the estimate
  - Variance of estimate

Methods of Estimation

- Holdout = Validation => 80/20
  - Reserve 2/3 for training and 1/3 for test

- Cross validation
  - Partition data into k disjoint subsets
  - k-fold: train on k-1 partitions, test on the remaining one
  - Leave-one-out: k=n

- Random subsampling
  - Repeated holdout

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Train, Test and Validate

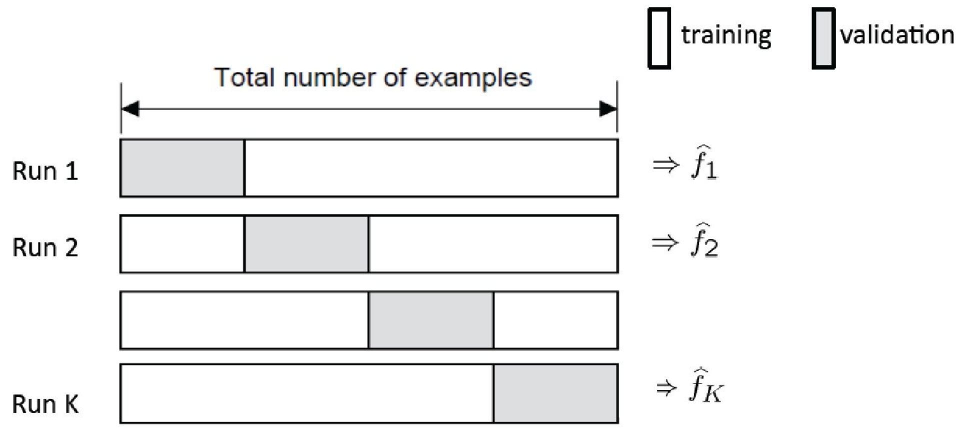
- **Training Set**: The actual dataset that we use to train the model (weights and biases in the case of Neural Network). The model sees and learns from this data.

- **Validation Set**: The sample of data used to provide an unbiased evaluation of a model fit on the training dataset while tuning model hyperparameters. The evaluation becomes more biased as skill on the validation dataset is incorporated into the model configuration.

- **Testing Set**: The sample of data used to provide an unbiased evaluation of a final model fit on the training dataset.

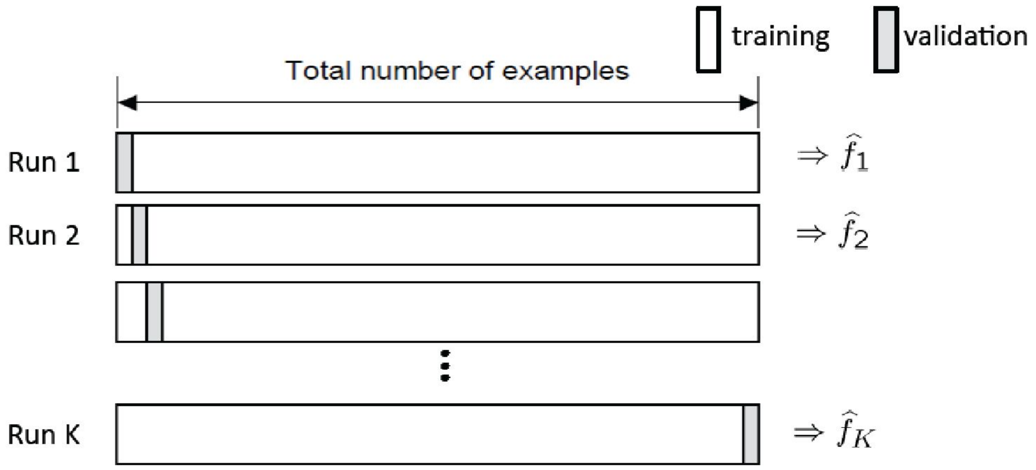
Cross Validation

- K-fold cross-validation
  - Create K-fold partition of the dataset.
  - Form K hold-out predictors, each time using one partition as validation and rest K-1 as training datasets.
  - Final predictor is average/majority vote over the K hold-out estimates.



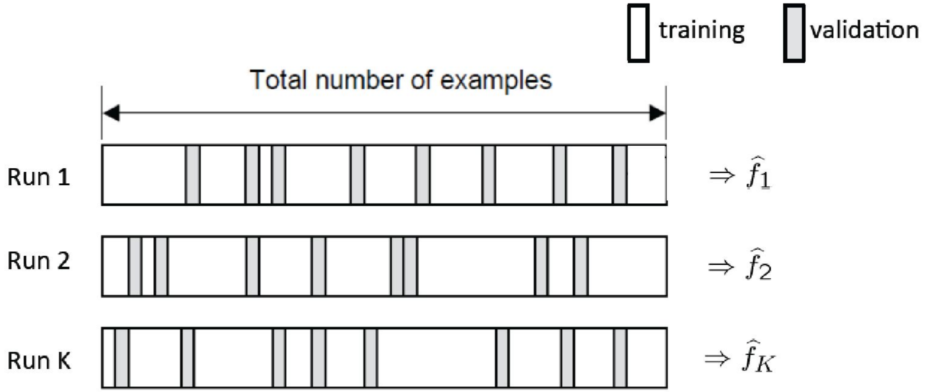
Cross Validation

- Leave-one-out (LOO) cross-validation
  - Special case of K-fold with K=n partitions
  - Equivalently, train on n-1 samples and validate on only one sample per run for n runs



Random Subsampling

- Randomly subsample a fixed fraction αn (0< α <1) of the dataset for validation.
- Train the model with remaining data as training data.
- Repeat K times
- Compute the average metrics for the K hold-out estimates.



What you should know?

- What is Generalization? Underfitting and overfitting
- Types of ensemble classifiers
  - Bagging (Random Forest)
  - Boosting
- How to model evaluation

Suggestion: If you're not very clear or want to learn more, please do read: "C. Bishop: Pattern Recognition and Machine Learning. Springer, 2006" (recommended text book). It will help you to understand the concept.

Acknowledgement

- *The following material have been referenced or partially used:*
  - *Lecture notes from Prof. Berrak Sisman and Prof. Zhao Na*
  - *National University of Singapore – Pattern Recognition Module (EE5907R)*
  - *National University of Singapore – Neural Networks Module (EE5904R)*
  - *University of Edinburgh, United Kingdom – Machine Learning And Pattern Recognition (MLPR, INFR11130)*
  - *Stanford University – Machine Learning (CS229)*